

Test Design

Simulation tests for yard planning and historical-replay acceptance tests for the re-handle metric

Deliverable 4 of 6 · Case Study no. 39 · PortFlow

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Case	PortFlow - Container Terminal Yard and Gate Operations		

1. Test Strategy

Testing is organised in layers. The two most important are **simulation-based tests** (does the yard planner behave well across many situations?) and **historical-replay acceptance tests** (does PortFlow meet the re-handle target on real data?).

- **Unit tests:** stacking rules, slot capacity checks, crane sequencing functions.
- **Integration tests:** planner ↔ broker ↔ execution controller; adapters against crane and gate simulators.
- **Simulation tests:** discrete-event model of vessels, trucks, cranes and stacks (section 2).
- **Acceptance tests:** historical replay against AC-0 and AC-1 (section 3).
- **Non-functional tests:** failover, load, security, usability.

2. Simulation-Based Tests for Yard Planning

A discrete-event simulation generates vessel arrivals, container flows and truck appointments. The yard planner under test is plugged into it, and results are compared with a baseline run of the legacy stacking rules. Each scenario is run with about 30 random seeds, and the mean and spread are reported.

ID	Scenario	Setup	Pass condition
SIM-1	Normal week	Typical vessel and truck volumes	Re-handle ratio ≤ 1.25
SIM-2	Peak load	+30% volume, vessels bunched together	Ratio ≤ 1.35 ; no stack-limit violation
SIM-3	Vessel delayed 12 h	ETD changes after stacking	Re-plan ≤ 5 min; blocked containers flagged
SIM-4	Crane breakdown	One crane fails mid-shift	Sequence re-optimised; no lost jobs
SIM-5	Mixed cargo	Reefer and hazardous containers	100% constraint compliance
SIM-6	Carrier no-shows	20% of booked trucks do not arrive	Gate queue within agreed limit
SIM-7	Planning engine killed	Planner stopped during a run	Execution continues; zero lost moves

3. Acceptance Tests Using Historical Replay

Why replay is stronger than a synthetic benchmark: it uses the terminal's real arrival patterns, real vessel delays and real container mixes, so the planner cannot be tuned to an invented test.

3.1 Procedure

Step	Action
1	Extract at least 3 months of real data (including a peak period): vessel schedules, gate-in/out events, yard moves, container departure times.
2	Clean and validate the data; record any gaps and how they were treated.
3	Calibrate (AC-0) : replay the events using the <i>legacy</i> stacking rules. The ratio must be 1.8 ± 0.1 . If not, fix the replay model first.
4	Run PortFlow on the same event stream: same containers, same departure times, no knowledge of the future beyond what the real planner would have.
5	Compute the re-handle ratio (section 4) overall, per week, per container type.
6	Decide (AC-1) : overall ≤ 1.25 and peak week ≤ 1.35 → PASS. Otherwise FAIL and analyse the 20 worst stacks.

4. Re-handle Measurement

Re-handle ratio = total crane lifts on retrievals ÷ containers retrieved

- A **lift** is counted each time a container is physically picked up in the yard for a retrieval job (including moving a blocking container).
- A container retrieved with no blockers contributes 1.0; one with a blocker contributes 2.0 (blocker move + own move).
- Measured for outbound containers (to vessel or gate) over the replay period, reported overall and by week.
- Data source: crane move logs from the replay, not manual estimates.

Worked example

Over a week, 1,000 containers are retrieved and the cranes make 1,800 lifts: ratio = 1,800 / 1,000 = **1.80** (current situation). PortFlow must bring the same week to at most 1,250 lifts: ratio ≤ **1.25**.

5. Other Test Cases

ID	Test	Expected result
FT-1	Stack a container with earlier ETD below one with later ETD (YA-2)	Planner rejects or avoids the placement
FT-2	Book a slot when yard capacity is full (SA-2)	Alternative windows offered; no overbooking
FT-3	Truck arrives 40 min late (SA-4)	Grace rule applied or rebooking offered
FT-4	ETD change on 50 containers (YA-4)	Re-plan within 5 minutes; blocked list produced
NT-1	Kill one execution node	Standby takes over; no gate or crane interruption
NT-2	Cut link to data centre (EX-2)	Terminal runs from plan cache
NT-3	Carrier A requests Carrier B's booking (NFR-5)	Access denied and logged
NT-4	500 concurrent slot bookings	Response within agreed time; no double-booking
UAT-1	Planners and gate staff run a pilot shift	Sign-off by terminal management

6. Entry and Exit Criteria

- **Entry**: SRS baselined; replay dataset cleaned; simulator calibrated (AC-0 passed).
- **Exit**: all Must requirements tested; AC-1 passed; no open critical defects; failover tests passed.